# MUHAMMAD ASIM IQBAL

*Data Science Graduate | Machine Learning & AI Enthusiast*

Nationality: Pakistani | Date of Birth: 17/02/2002 | Place of Birth: Multan, Pakistan | Gender: Male

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## CAREER & RESEARCH OBJECTIVE

A highly motivated Data Science graduate with a strong academic record and hands-on experience in machine learning, computer vision, automation, and data analytics. Demonstrated ability to design and deploy end-to-end AI solutions, ranging from biometric classification systems to business intelligence dashboards. Seeking admission to a postgraduate program in China under the Chinese Government Scholarship (CSC) / ANSO Scholarship to pursue advanced research in Data Science, Artificial Intelligence, and Machine Learning, with the long-term goal of contributing to innovative, data-driven solutions for real-world problems and strengthening academic and technological collaboration between Pakistan and China.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, JavaScript, HTML, CSS

**Web Development:** Flask, Streamlit, WordPress

**Databases:** MySQL, SQLite, Google Sheets Integration

**Data Science & Machine Learning:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn, Plotly

**Image Processing & Computer Vision:** OpenCV, CNN, Autoencoders (applied to fingerprint analysis and blood group prediction)

**Automation & Scripting:** Python Automation, Web Scraping, Google Apps Script

**Tools & Platforms:** Jupyter Notebook, Google Colab, Git, VS Code, Kaggle, Power BI, Tableau

**Soft Skills:** IT Support, ERP System Handling, Problem Solving, Cross-functional Communication

## EDUCATION

### BS Data Science

*Muhammad Nawaz Shareef University of Agriculture, Multan, Pakistan | Sep 2021 – July 2025*

## RESEARCH WORK

"Fingerprint-Based Blood Group Classification: A Biometric Approach to Blood Typing and its Potential Applications in Healthcare" — Independent research applying deep learning (CNN) and biometric image processing to develop a non-invasive method for blood group prediction, with direct relevance to healthcare diagnostics and biomedical AI applications.

## WORK EXPERIENCE

### Data Science Researcher — Fingerprint-Based Blood Group Prediction

*Independent Research Project | Feb 2025 – June 2025*

- Designed and trained a Convolutional Neural Network (CNN) to classify fingerprint patterns (whorl, arch, loop) and predict blood groups
- Preprocessed 1,000+ fingerprint images using OpenCV techniques including normalization, enhancement, and filtering
- Applied autoencoders for anomaly detection using reconstruction error thresholds
- Visualized classification results and performance metrics using confusion matrices and an interactive Streamlit dashboard

### IT Support Student / Studentship

*MNS University of Agriculture, Multan | Mar 2023 – Feb 2025*

- Provided IT support for faculty and staff, including troubleshooting hardware, software, and connectivity issues
- Managed and maintained ERP systems for student enrollment, course management, and exam results
- Resolved system-related issues efficiently through clear communication and collaborative teamwork

### Accounts and Finance Officer

*HMD Poultry Solutions | Nov 2024 – Aug 2025*

- Managed daily accounts operations including billing, payments, and expense tracking
- Created and maintained invoices and vouchers using Google Sheets and Apps Script
- Maintained accurate, up-to-date financial records for clients and the company

- Built dashboards to track payment status and generate financial reports

**Accounts and Finance Officer**
*Nutrion | Sep 2025 – Present*

- Manage daily accounts tasks including billing, payments, taxation, and expense tracking
- Create and manage invoices and vouchers using Google Sheets and Apps Script
- Develop custom software solutions to manage daily accounts and taxation with accuracy

**Data Analytics Projects — University Administration**
*Muhammad Nawaz Shareef University of Agriculture, Multan*

- Collected, cleaned, and transformed institutional datasets for academic and administrative insights
- Built Power BI dashboards to visualize student enrollment, faculty workload, and academic performance
- Conducted data analytics on student records to uncover trends and support data-driven decisions
- Presented actionable insights through interactive reports and charts for university stakeholders

## KEY PROJECTS

**TV Show Popularity Analysis**
*Oct 2024 – Mar 2025*

- Conducted an end-to-end analysis project using Python, including data scraping, cleaning, and exploratory data analysis
- Analyzed viewership trends, audience ratings, and social media sentiment to identify key factors driving show popularity
- Applied Pandas, Matplotlib, and Scikit-learn for data visualization and predictive modeling

**Diabetes Prediction Model**
*Feb 2024 – Aug 2024*

- Developed a machine learning model to classify diabetic and non-diabetic cases based on medical data
- Applied data preprocessing, feature selection, and supervised learning using Pandas, Scikit-learn, and Matplotlib
- Evaluated model performance using accuracy, precision, recall, and confusion matrix analysis

**House Price Prediction Model**
*Feb 2024 – Aug 2024*

- Built a regression-based model to estimate property prices from features such as location, size, and amenities
- Utilized NumPy, Scikit-learn, and Seaborn for data analysis, visualization, and model training
- Improved model accuracy through feature engineering, outlier treatment, and hyperparameter tuning

## PUBLICATIONS

Haq, E., Rajwana, M. A., Iqbal, M. A., & Kashif, M. (2025). Emerging Algorithms Reshaping Computer Vision: The Next Frontier in Visual Recognition. Journal of Statistics, Computing and Interdisciplinary Research, 7(2), 387–410.

- Co-authored a review paper evaluating emerging computer vision algorithms including Vision Transformers (ViTs), GANs, diffusion models, and hybrid computational paradigms, based on a systematic literature review of 50 curated papers

Rehman, M. A. U., Sarfraz, Z., Raffay, A., Rao, A. A., Iqbal, M. A., & Rehman, A. U. (2025). Enhancing QR Code Security Using a Hybrid Multi-Layer Steganography Encryption Model. Journal of Statistics, Computing and Interdisciplinary Research, 7(2), 377–385.

- Co-authored a research paper proposing a hybrid image steganography model combining Least Significant Bit (LSB) technique and Discrete Wavelet Transform (DWT) with 3D barcode and cloud-based management tools to enhance QR code data security

## CERTIFICATIONS & ONLINE COURSES

- **Get Started with Tableau:** https://coursera.org/share/d73dc70bdb843c0b5ae62af924820c29
- **Prepare, Clean, Transform, and Load Data using Power BI:** https://coursera.org/share/13090edce5ca5ee0ed7b8dfedc7278f4
- **Introduction to Relational Database and SQL:** https://coursera.org/share/d92d4d132f8eed081c93ec56de57fa31
- **Create Digital Marketing Campaign Dashboards in Tableau:** https://coursera.org/share/2dda60ec7a6964b2e38e35e07db375ce
- **Foundations of Data Science:** https://coursera.org/share/97e7d02cea2443dd0a3780ea095596bf

- **Application of Data Analysis in Business with R Programming:** https://coursera.org/share/6a6418408432aac3e840115bb7a1b9bd
- **System Administration and IT Infrastructure Services:** https://coursera.org/share/827baa42eeaccfafadc4c9b0f62cd49b
- **Google Advanced Data Analytics Capstone:** https://coursera.org/share/07603b7bdda97369b207582ade7ca305
- **Create a Landing Page using Mailchimp:** https://coursera.org/share/081385b8938fcb3570beae0773ccd95b
- **Simple Linear Regression for the Absolute Beginner:** https://coursera.org/share/3d090aa76371abf6b284e639c9553a60
- **Create and Manage a Project Dashboard with Wrike:** https://coursera.org/share/494c720489ec1a8f5711870bdab99cc0